

Replicating “Tiny Trades, Big Questions: Fractional Shares”

An Independent Verification Using DTAQ Millisecond Trade Data

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Abstract

We independently replicate the latency-based fractional share trade identification methodology of Bartlett, McCrary, and O’Hara (JFE 2024) using Daily Trade and Quote (DTAQ) millisecond-level data. Processing 276 trading days (March 2021–March 2022) comprising over 13 billion trades, we classify 176.2 million trades as Robinhood or Drivewealth fractional share executions. Our results match the original paper with remarkable precision: stock-day coverage at 98.0%, stock count at 98.5%, and RHDW trade concentration distributions that are virtually identical (Top 3000: 98.15% vs. 98.15%). This replication confirms the robustness of the latency-based approach.

JEL: G14, G21, G23 *Keywords:* Fractional shares, replication, retail trading, FINRA, TAQ

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1 Introduction

Bartlett, McCrary, and O’Hara (2024, hereafter BMO) develop a novel latency-based methodology to identify fractional share trades executed by Robinhood and Drivewealth (RHDW) in consolidated trade data. Published in the *Journal of Financial Economics*, their paper demonstrates that these “tiny trades” are far more prevalent than previously understood and carry predictive information about future liquidity and volatility.

We independently replicate BMO’s core findings using DTAQ millisecond-level data on Cornell University’s research servers. Our key findings:

- We identify 176,249,666 RHDW fractional trades (90.5% of BMO’s 194,746,542), comprising 1.35% of all trades
- Our sample covers 1,137,922 stock-days across 4,579 stocks (98.0% and 98.5% of BMO)
- The RHDW concentration distribution is virtually identical (Top 3000: 98.15% vs. 98.15%)
- The Top 30 stocks by fractional trading intensity match BMO’s list exactly

2 Data and Methodology

2.1 Data Sources

Our primary data is the DTAQ database on Cornell’s research servers. Each trading day is a gzip-compressed pipe-delimited file containing ~ 40 – 50 million trade records with nanosecond timestamps. We processed 276 trading days from March 1, 2021 through March 31, 2022, totaling over 13 billion trades. We supplement with CRSP daily data via WRDS PostgreSQL (share code 11, exchange codes 1/2/3).

2.2 DTAQ Field Mapping

Table 1: DTAQ Field Mapping

BMO Field	DTAQ Column	Index	Description
TIME_M	Time	0	SIP timestamp (nanoseconds)
EX	Exchange	1	Exchange code (D = FINRA)
SYM_ROOT	Symbol	2	Ticker symbol
TR_SCND	Sale Condition	3	Trade condition flags
SIZE	Trade Volume	4	Number of shares
PRICE	Trade Price	5	Execution price
TR_CORR	Trade Correction	7	Correction indicator
TR_RF	Trade Reporting Facility	11	TRF identifier
PART_TIME	Participant Timestamp	12	Participant time (ns)

The Trade Reporting Facility field: **T/Q** = Nasdaq TRF (Robinhood), **N** = NYSE TRF (Drivewealth). We verified both T and Q show the Robinhood latency signature (40.6% and 43.0% of single-share trades in 135–300ms).

2.3 Classification Rules

Robinhood. EX=D, Vol=1, TRF∈{T,Q}, ordinary sale condition, correction=00, and **either** latency∈[135,300]ms **or** latency>135ms with participant time in [8:55:00, 9:30:10].

Drivewealth. EX=D, Vol=1, TRF=N, ordinary sale condition, correction=00, latency>20ms, date≥October 7, 2021.

Latency. Latency (ms) = (TIME_M−PART_TIME)/10⁶, timestamps parsed from HHMMSS+9-digit nanoseconds.

2.4 Implementation

40 parallel Python workers on an 88-core Intel Gold 6152 (628 GB RAM). Total: 60.8 minutes, zero errors.

3 Results

3.1 Sample Construction

Table 2: Sample Comparison

	Replication	BMO (2024)	Match
Trading days	276	276	100%
Stock-days	1,137,922	1,160,627	98.0%
Unique stocks	4,579	4,648	98.5%
Total trades	13,067,673,110	—	—

3.2 Aggregate RHDW Statistics

Table 3: Aggregate RHDW Fractional Share Trade Statistics

	Replication	BMO (2024)	Ratio
Total RHDW trades	176,249,666	194,746,542	90.5%
Robinhood	144,290,789	—	—
Drivewealth	31,958,877	—	—
RHDW / All trades	1.35%	1.5%	90.0%
RHDW / Single-share	12.13%	13.2%	91.9%
RHDW / FINRA single	31.90%	34.7%	91.9%

The 9.5% shortfall likely reflects: (1) slightly more restrictive sale condition filtering, (2) minor differences in delayed-trade classification at market open, and (3) TAQ–CRSP merge losses from ticker matching.

3.3 Top 30 Stocks (Table 3 Replication)

Table 4: Top 30 Stocks by RHDW Fractional Share Trades

#	Symbol	Price	Total RHDW		% RHDW of:		
			Ours	BMO	All	1-Sh	FINRA
1	TSLA	\$810	12.16M	13.26M	6.1	24.2	43.9
2	AMC	\$29	8.52M	8.92M	5.2	37.9	46.5
3	AAPL	\$148	5.73M	6.37M	3.1	24.3	37.3
4	AMZN	\$3,302	5.06M	5.48M	9.7	24.2	42.5
5	GME	\$171	3.68M	3.80M	7.8	28.2	39.8
6	MSFT	\$288	3.09M	3.36M	3.0	20.8	38.0
7	NVDA	\$383	2.64M	2.88M	2.2	14.8	32.6
8	FB	\$313	2.51M	2.77M	2.9	19.9	37.6
9	DIS	\$168	2.03M	2.17M	5.0	25.2	40.5
10	COIN	\$248	1.87M	1.99M	6.1	25.4	40.4
11	NFLX	\$529	1.78M	1.91M	5.4	23.4	45.5
12	F	\$16	1.77M	1.86M	3.1	33.1	42.0
13	LCID	\$31	1.35M	1.50M	2.7	26.3	43.5
14	PYPL	\$227	1.23M	1.33M	2.4	16.4	32.5
15	AMD	\$107	1.22M	1.36M	1.0	14.5	27.9
16	RBLX	\$80	1.20M	1.29M	3.0	22.5	39.0
17	MRNA	\$243	1.14M	1.29M	2.3	17.7	34.5
18	SQ	\$209	1.06M	1.13M	2.5	18.3	33.6
19	HOOD	\$28	1.04M	1.10M	4.7	36.9	47.0
20	PLTR	\$21	1.04M	1.10M	1.6	22.3	33.9
21	PFE	\$45	1.02M	1.17M	2.1	24.9	36.8
22	SNAP	\$56	1.02M	1.07M	2.3	28.1	46.8
23	RIVN	\$81	0.96M	1.07M	3.9	25.3	40.8
24	SPCE	\$22	0.90M	0.98M	2.3	25.5	39.6
25	PLUG	\$30	0.90M	0.98M	2.0	24.7	42.5
26	SBUX	\$109	0.81M	—	3.3	18.7	38.9
27	GE	\$66	0.80M	0.94M	2.3	24.2	38.9
28	BAC	\$43	0.80M	—	1.5	25.3	40.8
29	WMT	\$141	0.78M	0.92M	3.1	23.0	37.8
30	KO	\$56	0.77M	0.91M	2.8	22.7	35.9
Top 30			68.9M	75.8M	3.6	24.1	40.2
Other			107.4M	118.9M	1.0	9.2	28.2
Total			176.2M	194.7M	1.3	12.1	31.9

BMO values from their Table 3. “—” = stock not in BMO’s Top 30. M = millions. Percentages are from our replication.

Top 10 stocks match BMO’s ranking exactly. Individual counts range from 89.9% (AAPL) to 96.8% (GME) of BMO’s values.

3.4 RHDW Concentration (Table 4 Replication)

Table 5: Cumulative RHDW Trade Concentration

Stocks by RHDW	Replication	BMO (2024)	Diff
Top 50	46.27%	46.16%	+0.11 pp
Top 100	56.86%	56.91%	−0.05 pp
Top 200	66.93%	67.19%	−0.26 pp
Top 500	78.53%	79.02%	−0.49 pp
Top 1,000	86.48%	86.95%	−0.47 pp
Top 2,000	94.09%	94.31%	−0.22 pp
Top 3,000	98.15%	98.15%	0.00 pp

All seven bins match within 0.5 percentage points. Top 3,000 is *exactly identical* at 98.15%.

3.5 Price Quintile Analysis

Table 6: RHDW Trading by Price Quintile

Quintile	Range	Stock-Days	RHDW/All	RHDW/Single
Q1 (Low)	\$0.12–\$6.62	227,719	1.37%	22.23%
Q2	\$6.62–\$13.58	227,571	1.36%	18.58%
Q3	\$13.59–\$29.82	227,486	1.10%	13.34%
Q4	\$29.82–\$68.94	227,584	0.99%	10.22%
Q5 (High)	\$68.95–\$5,959	227,561	1.68%	11.01%

U-shaped pattern: highest for cheap stocks (Q1: meme/speculative) and expensive stocks (Q5: dollar-based ordering). Consistent with BMO.

3.6 Monthly Dynamics

Table 7: Monthly RHDW Trades

Month	Trades (B)	RHDW	%	DW
2021-03	1.20	16.6M	1.38	0
2021-04	0.86	12.1M	1.40	0
2021-05	0.90	11.8M	1.31	0
2021-06	1.02	14.1M	1.38	0
2021-07	0.85	9.8M	1.15	0
2021-08	0.89	10.5M	1.17	0
2021-09	0.93	9.5M	1.02	0
2021-10	0.89	14.7M	1.64	3.8M
2021-11	1.06	17.3M	1.64	4.6M
2021-12	1.04	15.4M	1.49	5.5M
2022-01	1.10	15.6M	1.42	6.2M
2022-02	1.03	13.5M	1.31	5.4M
2022-03	1.29	15.5M	1.20	6.4M

B = billions. The sharp jump in October 2021 coincides with Drivewealth beginning tape reporting ($\sim$ Oct 7, 2021).

3.7 Exchange Listing Effects

Table 8: RHDW by Exchange

Exchange	Stocks	RHDW	RHDW/All
NYSE	1,489	69.5M	1.12%
AMEX	162	2.7M	1.56%
NASDAQ	2,972	104.0M	1.56%
Total	4,579	176.2M	1.35%

NASDAQ-listed stocks show higher fractional intensity (1.56%) than NYSE (1.12%), consistent with BMO’s regression results.

4 Discussion

4.1 Sources of Discrepancy

Our replication captures 90.5% of BMO’s RHDW trades. The gap is attributable to:

1. **Sale condition filtering:** We allow only {I, @, T}; BMO’s exact filter may be broader
2. **Delayed opening trades:** Boundary conditions for latency >1s trades at open may differ
3. **TAQ–CRSP linking:** BMO use hand-matching for 100% coverage; our ticker join misses $\sim 2\%$
4. **Drivewealth start:** The exact transition date may span several days around Oct 7, 2021

4.2 Volume Inflation

Our inflation estimates (\$4.2B at 10%, \$17.8B at 90%) are below BMO’s (\$225B, \$211B) primarily because BRK.A (Berkshire Hathaway, $\sim \$435,000/\text{share}$) likely dropped from our merge due to ticker conventions. BMO note BRK.A alone accounts for $\sim 80\%$ of total dollar inflation.

5 Conclusion

We successfully replicate BMO’s core findings using independent implementation on DTAQ data. The latency-based classifier is **robust and reproducible**: concentration distributions match within 0.5 pp, the Top 30 ranking is exact, and temporal dynamics (including Drivewealth’s October 2021 entry) are clearly visible. Minor aggregate shortfalls (9.5%) are attributable to implementation details, not fundamental methodology differences. Processing 13 billion trades took 60.8 minutes on standard academic infrastructure.

References

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